Subpixel-Pixel-Superpixel-Based Multiview Active Learning for Hyperspectral Images Classification

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***Abstract*— Active learning (AL) attempts to actively select the most representative or useful training samples in an iterative manner. The aim is to simultaneously improve the classification performance and reduce the manual labeling effort. In this article, a novel subpixel-pixel-superpixel-based multiview AL (MAL) (SPS-MAL) method is proposed for hyperspectral image (HSI) classification. Here, the multiple views are generated via extracting the subpixel-level, pixel-level, and superpixel-level information. The multiple views can reflect various characteristics of HSI, i.e., spectral mixture, spectral discrimination, and spectral–spatial structure. Therefore, the joint use of diverse and complementary information in multiple views will contribute to a better identification ability of different classes. In addition, a coarse-to-fine MAL algorithm is introduced to effectively select the most representative samples with the most uncertainty. Specifically, a disagreement analysis on multiple views and joint posterior probability estimation is used to query unlabeled samples. Along with the expansion of training samples, view-specific confidence scores are estimated to adaptively integrate the classification results of multiple views, according to their discrimination performance. In this way, the classification accuracy will be further boosted while the number of necessary training samples can be significantly reduced. The experimental classification results on three well-known HSIs demonstrate the effectiveness of the proposed SPS-MAL method.**

***Index Terms*— Active learning (AL), classification, hyperspectral image (HSI), multiview learning.**

I. Introduction

HYPERSPECTRAL image (HSI) classification is an active research field in remote sensing community, which is very important in different civil and military applications [1]. However, there still exist challenges in hyperspectral classification. For example, the mixture of spectral, the increase of intraclass diversities and the decrease of interclass diversities make it difficult to correctly identify different classes of land-covers. Nowadays, one of the most active and cutting-edge areas of hyperspectral classification is the deep learning (DL)-based classification methods. Compared with traditional methods, DL takes advantage of the ability to extract hidden and sophisticated structures in row data and the large flexibility on architectures [2]. Many researchers have put much attention on DL-based classification and achieved satisfying results [3]–[6]. In [3], the recurrent neural network (RNN) was first introduced for hyperspectral classification, which analyzes pixels as sequential data. In [4], the authors constructed the pixel pairs with center pixel and its surrounding pixels to ensure the advantages of convolutional neural networks (CNNs). In [5], the CNN combined with capsule networks enables higher classification accuracy while reducing the network complexity. Song *et al.* [6] fused the multiple-layer features to make use of the strong complementary and correlated information among different hierarchical layers. However, the generation of abundant labeled training samples is always a necessary requirement to obtain an accurate classification result, which is labor costly and time-consuming. Therefore, it is of great importance to investigate novel classification methods to improve classification performance while minimize the necessary training samples.

For the last decades, various active learning (AL)-based HSI classification methods [7]–[10] have been proposed to achieve better classification performance with the least labeling effort. Specifically, given a small set of initial labeled samples and a large set of unlabeled samples, AL attempts to iteratively expand the labeled samples by manually labeling a few carefully selected unlabeled samples. By selecting unlabeled samples in a smart way, instead of the traditional random manner, the classifier can be better trained for a higher discrimination capability. Moreover, the labeling cost will be reduced due to the decrease of necessary training samples.

In general, the AL methods can be divided into three categories: large margin-based heuristics, posterior probability (PP)-based heuristics, and committee-based heuristics [7]. For the large margin-based methods, margin sampling strategy is usually adopted to be combined with a support vector machine (SVM) classifier [11]–[13]. Tuia *et al.* [11] proposed a margin sampling-based AL method by considering the distribution in the feature space of the candidates and an entropy-based query by bagging algorithm. Later, Demir *et al.* [12] introduced a novel batch-mode AL approach by considering both the uncertainty and diversity of data when selecting the most informative samples. Pasolli *et al.* [13] combined the feature extraction process and large-margin nearest-neighbor-based AL steps into a unique framework. For

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the PPs-based methods, the PPs are used as an acceptable measurement to guide the selection of samples with maximum uncertainty. This is reasonable due to that PPs can reflect the confidence of the estimated class label of a sample. In [14], a commonly used AL strategy of posterior probability-based heuristics, named breaking ties (BTs), was introduced to select the most uncertain samples for classifiers. Then, the BT-based AL strategy is modified in [15] to better balance uncertainty and diversity during sample selection. The AL methods by committee-based heuristics utilize the disagreement of different classifiers to distinguish samples that are difficult to be classified correctly. In [16], the unlabeled samples whose labels vary before and after the Markov random field (MRF) process are picked out as sample candidates for further selection. In [17], a random walker-based classifier and extended random walker-based classifier are integrated as the committee members to select the candidate samples, which are further filtered by the modified BT strategy. In addition, based on the assumption that the decision boundary between classes has to lie in the low-density regions of feature space, the clustering strategy is usually adopted to select informative samples in AL related works [18]–[20]. In [18], a cluster tree is constructed to avoid biased sampling. Patra and Bruzzone [19] utilized the output score of SVM to identify the low density of the kernel space, which is assumed to be the most ambiguous area. In [20], unsupervised learning-based clustering is naturally combined with the AL where points near class boundaries will be selected. In one of our previous work [21], the clustering strategy is also used to enhance the diversity property of the selected samples.

In machine learning community, multiview learning (MVL) has been demonstrated to be a more effective learning strategy in contrast to single-view learning. For the MVL, data are first transformed into disjoint views (i.e., feature subsets, attributes), each of which is assumed to be sufficient to learn the correct target concept. Then, the complementary information of multiple views can be jointly utilized to predict a more accurate labeling result [22]. Since multiple views can reveal the data characteristics from various aspects and provide diverse information for model learning, multiview learning has been explored extensively, including multiview supervised methods [23], multiview semisupervised methods [24], multiview AL (MAL)-based methods [25], multiview ensemble methods [26], and multiview transfer learning-based methods [27].

Recently, a few attempts have been done to introduce the MAL to solve remote sensing image classification tasks. In the MAL framework proposed in [28], multiple views are generated by utilizing spectral-correlation and spatial-similarity. Xu *et al.* [29] investigate the class prediction ability of each single view to adaptively combine labeling probabilities by different views. A MAL based on 3-D redundant wavelet transform is proposed in [30], where a singularity-based criterion considering spatial information is used to select more informative samples. In general, these MAL methods can effectively improve HSI classification performance due to the good properties of multiview learning. However, most of the multiple views are generated based on the pixelwise spectral feature or spectral–spatial features extracted within a local region of fixed size, which ignore the shape-adaptive structural properties and phenomenon of the spectral mixture. Based on the above analysis, it is necessary to investigate a new MAL method to further improve classification performance, by jointly considering the spectral discrimination, irregular structures of land-covers and spectral mixture characteristics. Thanks to the widely developed spectral unmixing techniques [31]–[33], it is possible to extract spectral mixture information at a subpixel level, which can contribute to the categorization of seriously mixed spectral pixels. In addition, the superpixel segmentation algorithms have been demonstrated to be effective in extracting shape-adaptive spatial structure information [34]–[36]. By incorporating the spatial context, the superpixel-level information can be used to smooth classification results in local homogeneous regions. In this article, a novel SPS-MAL method is proposed for HSI classification. First, the multiple views (subpixel-view, pixel-view, and superpixel-view) are generated via extracting the pixel-level spectral discrimination information, the subpixel-level spectral mixture information and the superpixel-level spatial structure information. Then, a coarse-to-fine AL strategy is introduced for selecting new samples and updating classification results in an iterative manner. In the coarse stage, disagreement analysis on multiple views is conducted, to initially generate a set of candidate samples, which are assumed to be difficult to be correctly distinguished. In the fine stage, the uncertainty of the candidate samples is measured according to the posterior probabilities, which are then used to query and select the most representative samples. With all the newly selected training samples, the final classification results are obtained via introducing confidence scores of each view to adaptively integrate the class estimations of multiple views. This work can be seen as a further extension of our previous works [35] for AL. Specifically, feature-level and decision-level fusion schemes were proposed in [35], where subpixel-level, pixel-level and superpixel-level characteristics are combined to boost supervised classification accuracy for HSIs. Here, we proposed an effective AL method via utilizing the complementary and diverse information provided by three levels, to further improve classification performance while reducing training samples. The focus of this article is to design a more effective sample selection rule and labeling strategy in an AL scheme, where disagreement analysis on multiple views and probabilistic fusion is introduced.

The remainder of this article is organized as follows. The proposed subpixel-pixel-superpixel based MAL (SPS-MAL) method for HSI classification is described in detail in Section II. Experimental results are given in Section III. Finally, discussions and future work are concluded in Section IV.

II. Proposed Method

Fig. 1 shows the schematic of the proposed SPS-MAL method, which mainly consists of two key steps, i.e., multiview generation and MAL. In the first step, the subpixel-view, pixel-view, and superpixel-view are generated to characterize HSI from different aspects. The second step aims at updating

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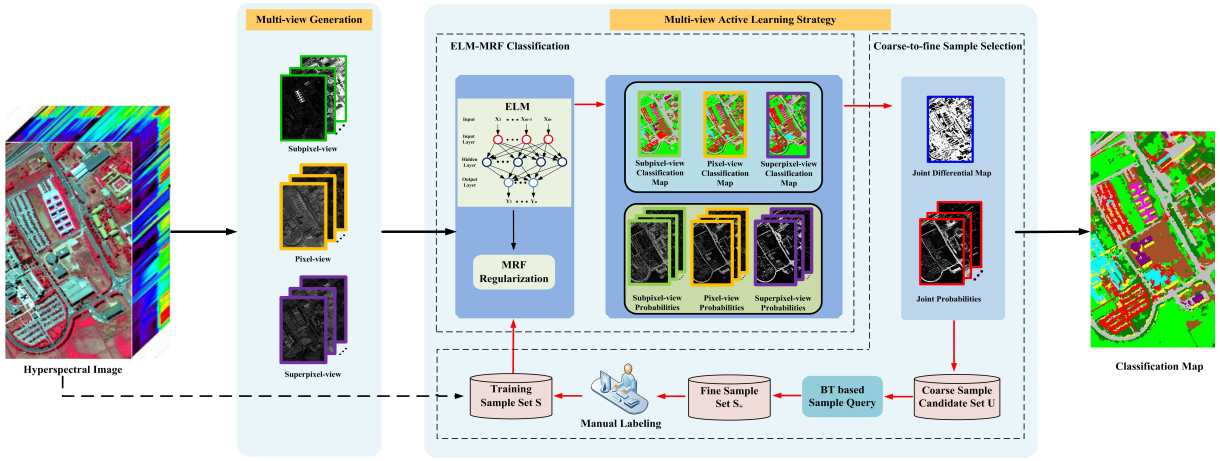


Fig. 1. Framework of the proposed SPS-MAL method for HSI classification.

both the training samples and classification results. In particular, this step is conducted in an iterative manner until the predefined number of training samples is obtained, and the final classification map can be generated via adaptive integration of multiview classification results in the last iterative time. Here, the process of coarse-to-fine sample selection is guided by disagreement analysis and joint posterior probability estimation on multiple views. Following our previous work in [21], the classification results and posterior probabilities are estimated by applying the extreme learning machine [37] and MRF (ELM-MRF) regularization. The procedure of the proposed SPS-MAL method is summarized in Algorithm 1 and the related descriptions are given below in detail.

A. Multiview Generation

The multiple views (including the pixel-view, subpixel-view, and superpixel-view) are generated from the HSI, reflecting the characters of different aspects.

1) Pixel-View: In the pixel-view, each pixel is characterized by the spectral vector in the data cube of HSI, providing discriminative spectral information. Suppose the original HSI is represented as $H \in R^{m\times n\times h}$, where $m \times n$ is the spatial size and h is the band number. Then, each sample in the pixel-view can be denoted as $\mathbf{z}_i^{\text{pix}} \in R^h$, where $i = 1, \ldots, m \times n$.

2) Subpixel-View: The subpixel-view is characterized by the abundance fractions with respect to endmembers. A well-known linear unmixing model [38] is adopted to describe the spectral mixture information

$$\mathbf{x}_i = \sum_{e=1}^{p_e} \Phi_i^e E_e + n_i \tag{1}$$

where E_e is the spectral response of endmember e, $\Phi_i = [\Phi_i^1, \Phi_i^2, \ldots, \Phi_i^{p_e}]$ is the corresponding fraction abundance of pixel $\mathbf{x}_i$, and n_i is the addictive noise item. According to [39], a minimum noise fraction-based dimensionality reduction is performed first, followed by the unsupervised mixture-tuned matched filtering method for the estimation of fractional abundances of spectral endmembers via the orthogonal subspace projection algorithm. In detail, given the number of endmembers, p_e, the K-means [40], as an unsupervised clustering algorithm, is used to divide all the pixels into p_e clusters. Then, the centroid corresponding to each cluster, denoted as E_e $(e = 1, 2, \ldots, p_e)$, can be considered as each "endmember." Therefore, the abundance fraction Φ_i of E_e for each specific $\mathbf{x}_i$ can be estimated by

$$\Phi_i^e = \left(\left(E_e^T \mathbf{R}^{-1} E_e\right)^{-1} \mathbf{R}^{-1} E_e\right)^T \mathbf{x}_i$$
$$\text{s.t. } \mathbf{R} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{x}_i \mathbf{x}_i^T \tag{2}$$

where $\mathbf{R}$ is the correlation matrix, N is the number of all pixels in HSI, $N = m \times n$. By (2), the fractional abundances, $\Phi_i = [\Phi_i^1, \ldots, \Phi_i^{p_e}]$ for each pixel $\mathbf{x}_i$ can be estimated. As a result, each sample in the subpixel-view, $\mathbf{z}_i^{\text{sub}}$, can be generated, $\mathbf{z}_i^{\text{sub}} = \Phi_i$.

3) Superpixel-View: Here, the entropy rate superpixel (ERS) [41] algorithm, as an effective oversegmentation method, is adopted to generate superpixels from HSI. Since the first three principal components (PCs) by the principal component analysis (PCA) contain the majority of structural and detailed information in the HSI, they are extracted first. Then, based on PCs, the ERS generates a subset of edges $E_s \in E$ by solving a graph optimization problem, denoted as

$$\max_{E_s}\{H(E_s) + \lambda F(E_s)\}, \quad \text{s.t. } E_s \in E \tag{3}$$

where H is the entropy rate of the random walk on graph G, and F is a balancing term to encourage clusters with similar sizes and reduce the number of unbalanced superpixel. The superpixel map can be generated according to the subset of edges E_s, which can segment the spatial area in HSI into multiple nonoverlapping homogeneous patches. Because of the spatial–spectral similarities within each superpixel S_i, it is reasonable to average the spectral to represent the common spectral–spatial information. The mathematical definition for this superpixel-view is denoted by (4), where M_i is the number of pixels within S_i

$$\mathbf{z}_i^{\text{sup}} = \frac{1}{M_i}\sum_{i\in S_i} \mathbf{x}_i. \tag{4}$$

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Algorithm 1 SPS-MAL

Input:
Subpixel-view, $\mathbf{Z}^{\text{sub}}$;
Pixel-view, $\mathbf{Z}^{\text{pix}}$;
Superpixel-view, $\mathbf{Z}^{\text{sup}}$;
A set of initial training samples, $S = S_0$;
A set of newly added training samples, $S_+ = \phi$;
Initial iteration time, $T_{iter} = 0$;
Iteration time, T;
Number of new labeled samples for each iteration, N_+.
Output:
Updated set of training samples, S;
Final classification result, $\hat{\mathbf{Y}}$;

Repeat:
1. $T_{iter} = T_{iter} + 1$, S=$S \cup S_+$;
2. ELM-MRF based classification. (i) Train the ELM classifiers with S and $\mathbf{Z}^{\text{pix}}$, $\mathbf{Z}^{\text{sub}}$, $\mathbf{Z}^{\text{sup}}$;
(ii) Estimate the class probabilities by (6), denoted as p^{pix}, p^{sub}, and p^{sup}.
(iii)Use the MRF to optimize classification maps by (7), denoted as $\mathbf{Y}^{\text{pix}}$, $\mathbf{Y}^{\text{sub}}$, and $\mathbf{Y}^{\text{sup}}$.
3. Coarse-to-fine Sample Selection.
Coarse stage:
(i) Use (8) to generate differential maps, denoted as: $D^{(pix,sub)}$, $D^{(sub,sup)}$, and $D^{(pix,sup)}$;
(ii) Generate joint differential map $D^{\text{sps}}$by (9);
(iii) Generate a set of candidate samples U by (10).
Fine stage:
(i) Estimate the view-specific adaptive weights by (11) and (12);
(ii) Calculate the joint posterior probabilities by (13);
(iii) Query the samples from U by (14), and manually label them
for the selection of N_+ new sample set S_+.
Until $T_{iter} > T$.
4. Estimation of final classification map.
(i)Generate the updated training sample set S;
(ii)Estimate the joint class probabilities by (13) and use the MRF to optimize classification result $\hat{\mathbf{Y}}$ by (7).

B. ELM-MRF Classification

1) ELM: ELM is a neural network with only one hidden layer and one linear output layer. Owing to the advantages of ELM in dealing with multiclass problem and its advantage in the convergence rate [37], it is selected as the basic classifier here. The weights between the input and the hidden layers are randomly assigned, and the weights of the output layer are computed using a least squares method [42]. For C classes, let the class labels be defined as $y_i \in \{1, -1\}$. Then, a constructed row vector $y_i = [y_i^1, y_i^2, \ldots, y_i^C]$ indicates the class to which a sample $\mathbf{x}_i$ belongs. The output function of an ELM with L hidden nodes can be expressed as

$$f(\mathbf{x}_i) = \sum_{l=1}^{L} \beta_l h(w_l \cdot \mathbf{x}_i + b_l) \tag{5}$$

where $h(\cdot)$ is a nonlinear activation function (e.g., sigmoidal function), β_l and w_l denote the weight vector connecting the lth hidden node to the output and input nodes, respectively, and b_l is the bias of the lth hidden node.

Recently, ELM has shown promising results for the classification of hyperspectral images (HSIs) [43], [44]. Therefore, it is used as the basic classifier in the proposed method. By (5), ELM-based decision output f for each test pixel, $\mathbf{x}_i$, is obtained, which can be used to estimate the class probability. Generally, the probability should be higher for a larger output of the decision function. Thus, the following form is adopted to estimate the probabilities:

$$p_q = \frac{1}{1 + \exp(Af + B)} \tag{6}$$

where p_q means the conditional class probability of the qth classifier (subpixel-view, pixel-view, and superpixel-view-based classifiers), and (A, B) are parameters estimated for ELM that can be estimated via the minimization of the cross-entropy error over the validation data.

2) MRF Regularization: Following our previous work [21], [34], it is suggested to refine the isolated wrong classification results via incorporating the MRF regularization. In the MRF-based framework, the task of labeling can be typically represented as a minimization of an energy function, denoted by

$$\hat{\mathbf{Y}} = \min \left\{ \sum_{\mathbf{x}_i \in \mathbf{X}} D_i(x_i) + \beta \sum_{i,j \in N_i} V_{ij}(y_i, y_j) \right\}. \tag{7}$$

In (7), the data item can be rewritten as $D_i(x_i) = -\log p_i$. The spatial item $V_{ij}(y_i, y_j)$ is calculated by $V_{ij}(y_i, y_j) = 1 - \delta(y_i, y_j)$, where $\delta(\cdot, \cdot)$ is the Kronecker delta function ($\delta(a, b) = 1$ if $a = b$, and $\delta(a, b) = 0$ otherwise). The parameter β is used to balance both the data item and the spatial item. By iteratively minimizing (7) with graph cut-based optimization algorithm[1] [45], the class labels can be obtained. In brief, the minimum cut is used to optimally select a set of image pixels to change their labels to a given label in each iteration. By this way, the value of energy cost function can be reduced gradually. Along with the iterations, the final classification result, $\hat{\mathbf{Y}}$, will be obtained until any change of the labels will no longer lead to further minimization of the energy cost function in (7). For more details about the solution to (7) in the context of HSI classification, refer to related works in [46]–[48].

C. Coarse-to-Fine Sample Selection

Here, a coarse-to-fine sample selection strategy is introduced for the selection of samples which are most representative and difficult to be distinguished.

1) Coarse Sample Selection: For the coarse state, unlabeled samples which have contrary classification estimations on different views are selected as the initial sample candidate. These samples are labeled differently by multiview-based

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decision makers, denoting that they are difficult to be correctly classified. By comparing any two of the classification results, Y^{pix}, Y^{sub}, and Y^{sup}, three differential maps are first generated, represented as $D^{(\text{pix,sub})}$, $D^{(\text{sub,sup})}$, and $D^{(\text{pix,sup})}$, respectively. The differential maps are used to indicate the labels of which pixels in the HSI are difficult to be correctly classified, according to the inconsistency between results based on different views. Here, the differential map is defined as

$$D^{(\text{pix,sub})}(\mathbf{x}_i) = \begin{cases} 0, & Y^{\text{pix}}(\mathbf{x}_i) = Y^{\text{sub}}(\mathbf{x}_i) \\ 1, & Y^{\text{pix}}(\mathbf{x}_i) \neq Y^{\text{sub}}(\mathbf{x}_i). \end{cases} \tag{8}$$

Then, the $D^{(\text{sub},P\text{sup})}$ and $D^{(\text{pix,sup})}$ are generated in a similar way using (8). Finally, the joint differential map, D^{sps}, is obtained according to the union set, denoted as

$$D^{\text{sps}} = D^{(\text{pix,sub})} \cup D^{(\text{pix, sup})} \cup D^{(\text{sub,sup})}. \tag{9}$$

Based on the differential map, D^{sps}, a candidate set of samples, U, can be generated by

$$U = \{q | q \in U_0, q \notin S\}, \quad \text{s.t. } U_0 = \{q | D^{\text{sps}}(\mathbf{x}_q) = 1\} \tag{10}$$

where S represents the labeled training sample set, which should be removed from U_0 to avoid the same samples are repeatedly selected.

2) Fine Sample Selection: For the fine stage, view-specific confidence score and the BT are integrated to query all the sample candidates according to uncertainty. First, view-specific confidence score is estimated, which are used for the generation of joint class probabilities p^{sps}. By training the ELM classifier corresponding to different views, with respect to the updated set of training samples S, the conditional probabilities can be estimated, denoted as p^{pix}, p^{sub}, and p^{sup}. The introduction of confidence score aims at adaptively assigning appropriate nonuniform scores to each classifier based on different views, according to the capability in discriminating various materials. Take the class probabilities by the subpixel-based estimation as an example, all the probabilities related to a pixel $\mathbf{x}_i$ are first sorted in a descending order, denoted as $\{p_{i,1}^{\text{sub}} \geq p_{i,2}^{\text{sub}} \geq \cdots \geq p_{i,C}^{\text{sub}}\}$. Here, $\{p_{i,k}^{\text{sub}}, k \in (1, 2, \ldots, C)\}$ represents the probability of $\mathbf{x}_i$ belonging to each class, which is estimated by the subpixel-level view-based classifier. Similar operations are also conducted on the pixel- and superpixel-based views to generate the sorted probability vectors, i.e., $\{p_{i,1}^{\text{pix}} \geq p_{i,2}^{\text{pix}} \geq \cdots \geq p_{i,C}^{\text{pix}}\}$ and $\{p_{i,1}^{\text{sup}} \geq p_{i,2}^{\text{sup}} \geq \cdots \geq p_{i,C}^{\text{sup}}\}$. Generally, if the first and second largest probabilities have a larger difference, the estimated label is more likely to be correct. This way, the weaker classifier relying on the less-discriminative feature will be assigned a lower weight, and vice versa. With the goal of obtaining more diversity in the composition of the training set, we define the confidence scores related with the subpixel-, pixel-, and superpixel-based views as α^{sub}, α^{pix}, and α^{sup}, denoted by

$$\begin{aligned} \alpha^{\text{sub}} &= \frac{1}{N_U} \sum_{i \in U} \left(p_{i,1}^{\text{sub}} - p_{i,2}^{\text{sub}} \right) \\ \alpha^{\text{pix}} &= \frac{1}{N_U} \sum_{i \in U} \left(p_{i,1}^{\text{pix}} - p_{i,2}^{\text{pix}} \right) \\ \alpha^{\text{sup}} &= \frac{1}{N_U} \sum_{i \in U} \left(p_{i,1}^{\text{sup}} - p_{i,2}^{\text{sup}} \right) \end{aligned} \tag{11}$$

where N_U is the number of samples in U. Thus, the weight of each classifier is calculated by

$$\begin{aligned} w^{\text{pix}} = \frac{\alpha^{\text{pix}}}{\alpha^{\text{sps}}}, \quad w^{\text{sub}} = \frac{\alpha^{\text{sub}}}{\alpha^{\text{sps}}}, \quad w^{\text{sup}} = \frac{\alpha^{\text{sup}}}{\alpha^{\text{sps}}} \\ \text{s.t. } \alpha^{\text{sps}} = \alpha^{\text{sub}} + \alpha^{\text{pix}} + \alpha^{\text{sup}}. \end{aligned} \tag{12}$$

Then, the joint posterior probabilities related to the ith pixel of multiview, p_i, can be calculated by

$$p_i^{\text{sps}} = w^{\text{sub}} p_i^{\text{sub}} + w^{\text{pix}} p_i^{\text{pix}} + w^{\text{sup}} p_i^{\text{sup}}. \tag{13}$$

Based on the generated joint probabilities, the BT-based query algorithm is used to select the most informative samples with most uncertainty for labeling in the fine stage. The BT is an active selecting strategy in AL [15]. Given the posterior probabilities of a set of samples, assume p_i is the posterior probability vector of sample q_i, where the components are sorted in a descending order, i.e., $p_{i,j} \geq p_{i,j+1}$, $j \in \{1, \ldots, C-1\}$, $\sum_j p_{i,j} = 1$. The BT criterion aims to select those samples with largest uncertainty

$$\begin{aligned} e_i &= p_{i,1} - p_{i,2} \\ q_i &= \min_i e_i, \quad i = 1, \ldots, n \end{aligned} \tag{14}$$

where n is the sample size, $p_{i,1}$, $p_{i,2}$ are the first and second largest items in a probability vector. According to (14), we can get the most uncertain samples, and in practice, we usually choose batch selection strategy, rather than select one sample per iteration. In the final iteration, the joint probabilities are utilized in (7) and (13), to get the final classification results.

III. Experimental Results and Discussion

In this section, the classification performance of the proposed SPS-MAL method is evaluated experimentally on three widely used hyperspectral remote sensing data sets, including "AVIRIS Indian Pines," "University of Pavia," and "Salinas." To demonstrate the effectiveness and superiority of the proposed method, four AL methods and two fusion-based supervised methods are compared numerically and visually. Among them, the AL-based methods include spectral–spatial LORSAL classifier with AL and multilevel logical spatial propagation (LORSAL-MLL) algorithm [15], the maximizer of the posterior marginal by loop belief propagation with AL (MPM-LBP) algorithm [49], the random-walker based collaborative learning (RWASL) algorithm [17], and the iterative clustering-based AL (ICAL) [21]. Regarding the compared AL methods, ICAL, and SPS-MAL share the same classifier, i.e., ELM-MRF. Besides, the ICAL, RWASL, and the proposed SPS-MAL method utilize ensemble learning strategy, where they all make use of the classification difference based on different classifiers. The difference is that ICAL takes the classification difference between adjacent iterations as a criterion to select candidate samples and RWASL compares the classification results by using the random-walker algorithm and extended random-walker algorithm to generate the candidate set. In this article, disagreement analysis on multiviews (subpixel-level, pixel-level, and superpixel-level views) combined with probability analysis is used for the most useful sample selection. In addition, apart from the

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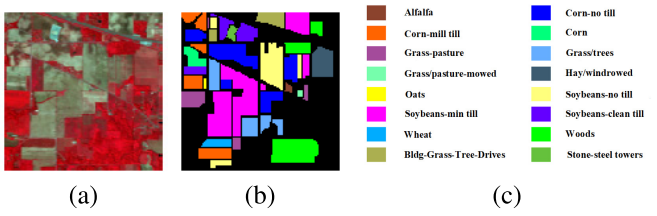


Fig. 2. Indian Pines image. (a) False color map. (b) Groundtruth. (c) Classes name.

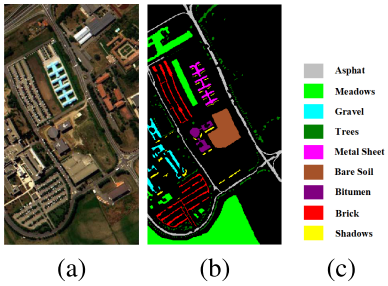


Fig. 3. University of Pavia image. (a) False color map. (b) Groundtruth. (c) Classes name.

ensemble learning strategy, RWASL adopts semisupervised learning to further boost classification accuracy. Among the two supervised methods, the multiscale covariance map feature extraction (MCMs) [50] adopts the deep features of HSIs and improve the classification performance via the integration of multiple layers of deep features, while the hierarchical guidance filtering (HiFi)-based classification method [51] introduces an ensemble framework to combine spectral and spatial information in different scales. In terms of objective metrics to measure classification performance, the overall accuracy (OA), average accuracy (AA), Kappa coefficient, and class accuracy (CA) values are calculated by averaging the results obtained in ten independent runs. Moreover, the visual comparison results of classification maps are also shown below.

A. Data Description

Three HSIs, including the "AVIRIS Indian Pines," "University of Pavia," and "Salinas," are used to test the performance of the proposed classification method. The corresponding false-color composite images and groundtruth to the three data sets are shown in Figs. 2–4, respectively.

1) AVIRIS Indian Pines: The "AVIRIS Indian Pines" image, which covers the agricultural Indian Pines test site in Northwestern Indiana, was collected by the Visible Infrared Imaging Spectrometer (AVIRIS) sensor. This image is of size 145 × 145 × 220, which has a spatial resolution of 20 m and a spectral range from 0.2 to 2.4 μm. Before the classification, 20 spectral bands (i.e., 104–108th, 150–163rd, and 220th) are discarded due to water absorption.

2) University of Pavia: The "University of Pavia" image was acquired by the ROSIS-03 sensor over the campus at the University of Pavia, Italy. It contains 103 spectral bands after discarding the seriously noisy bands, and each band is of size 610 × 340. The spatial resolution of this image is 1.3 m and the spectral coverage ranges from 0.43 to 0.86 μm.

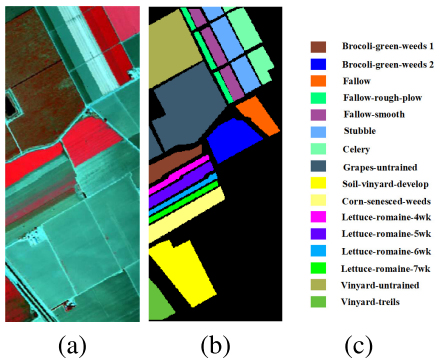


Fig. 4. Salinas image. (a) False color map. (b) Groundtruth. (c) Classes name.

3) Salinas: The "Salinas" image was also captured by the AVIRIS sensor over the area of Salinas Valley, California, USA, with a spatial resolution of 3.7 m/pixel. After discarding 20 water absorption bands, this image contains 224 bands of 512 × 217 size.

B. Parameter Discussion

For the proposed SPS-MAL method, there are two important parameters affecting our experimental performance that should be predefined. One is the smooth parameter β in the ELM-MRF classifier, and the other one is parameter N_c which determines the average number of pixels within each superpixel. Here, the effect of these two parameters is analyzed by evaluating the classification accuracies on two widely used test data sets, "University of Pavia" and "AVIRIS Indian Pines." For each test image, the number of initial training samples for each class is 3, which are randomly selected. For a fair comparison, all compared AL methods share the same initial training samples, and 15 more new samples are selected per iteration for these two images. Here, the values of parameter β ranges from 0 to 5, and values of parameter N_c ranges from 3 to 17 with interval 2. Fig. 5 presents the OA indices of two test data sets, along with varying parameters when iteration time is 1, 5, 10, and 15, respectively. In general, it can be observed that the classification accuracies are worse when the values of both β and N_c are small. In addition, along with the increase of parameter values for β and N_c, the OA curve exhibits a trend of first increasing and then decreasing. This can be explained as follows. The parameter β is used to balance both the data item and the spatial item. On the one hand, the introduction of β in (7) can help to correct isolated wrong label estimation results. On the other hand, a much larger value of β may lead to an oversmoothing classification map. Thus, an appropriate parameter value of β can contribute to higher overall classification accuracies. According to the above analysis, it is suggested to set the parameter β to be 3 for the three data sets. In terms of the parameter N_c, it is used to determine the number of superpixels. Let $m \times n$ represent the spatial size of test HSI, the predefined number of superpixel is set to be $m \times n/N_c^2$. Generally, a smaller value of N_c determines a larger number of superpixels, leading to the fact that each superpixel contains less pixels, and vice versa. However, a much larger N_c value

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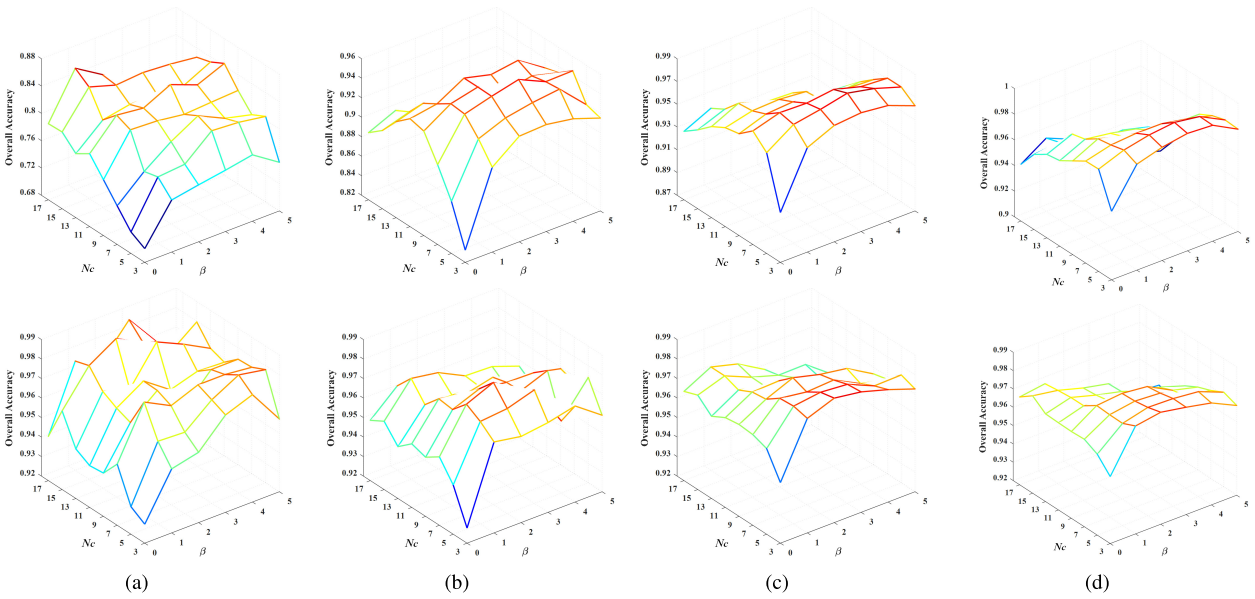


Fig. 5. Effect of parameters β and N_c on OAs. The figures in the first and second rows show the classification results of Indian Pines and University of Pavia, respectively. The results in each row from left to right correlated with the (a) 1st, (b) 5th, (c) 10th, and (d) 15th iteration times.

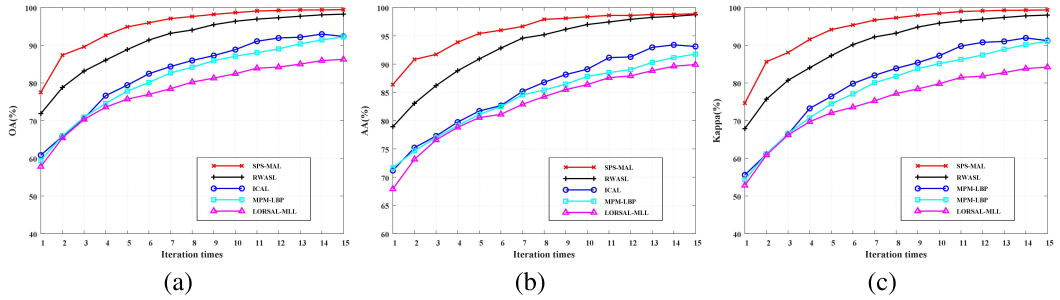


Fig. 6. Comparison classification results of Indian Pines image by different methods when the iteration times increase from 1 to 15, with 15 new samples selected per iteration. (a), (b), (c) are curves of OA, AA, and Kappa, respectively.

easily leads under segmentation. In addition, too much smaller value of N_c may cause that each superpixel cannot provide enough spatial information for expected classification. Therefore, an appropriate estimation for the number of superpixels is important for accurate classification. Based on the results in Fig. 5, it can be observed that the optimal parameter of N_c should be values between 5 and 7. Here, considering the image of "Salinas" contains more homogeneous regions of larger size than the other two images, we set parameter N_c as 5 for Indian Pines and University of Pavia and 7 for Salinas. Additionally, in the following experiment parts, the parameters of other methods for comparison are set according to their best performances.

C. Comparison Results by Different Classification Methods

Here, the classification results by the proposed SPS-MAL method are visually and quantitatively compared with both AL methods (LORSAL-MLL, MPM-LBP, ICAL, and RWASL), supervised methods (MCMs and HiFi). For the two supervised methods, a balanced number of training samples for each class are randomly selected. The total number of training samples for all the compared methods are the same, and the final results of all methods are calculated by averaging the results obtained in ten independent runs.

For the test image of Indian Pines, the classification performance of different AL-based methods are compared first. Fig. 6 shows the changing curves of OA, AA, and Kappa coefficients obtained by different AL methods with iteration times varying from 1 to 15, and in each iteration, 15 samples will be selected. In general, the LORSAL-MLL, MPM-LBP, and ICAL methods present competitive performance in terms of objective metrics, while there is an obvious gap compared to the RWASL and the proposed SPS-MAL method. Another obvious observation is that the difference of classification accuracies between various compared methods and the proposed SPS-MAL method is decreasing gradually, along with the increase of iterations. In particular, the RWASL method generates competitive classification results for the last few iteration times, owning to the semisupervised learning process of jointly exploiting both the labeled samples and a few sets of unlabeled samples. Overall, the proposed SPS-MAL method continuously obtains higher classification accuracies among the compared several AL approaches. This should be

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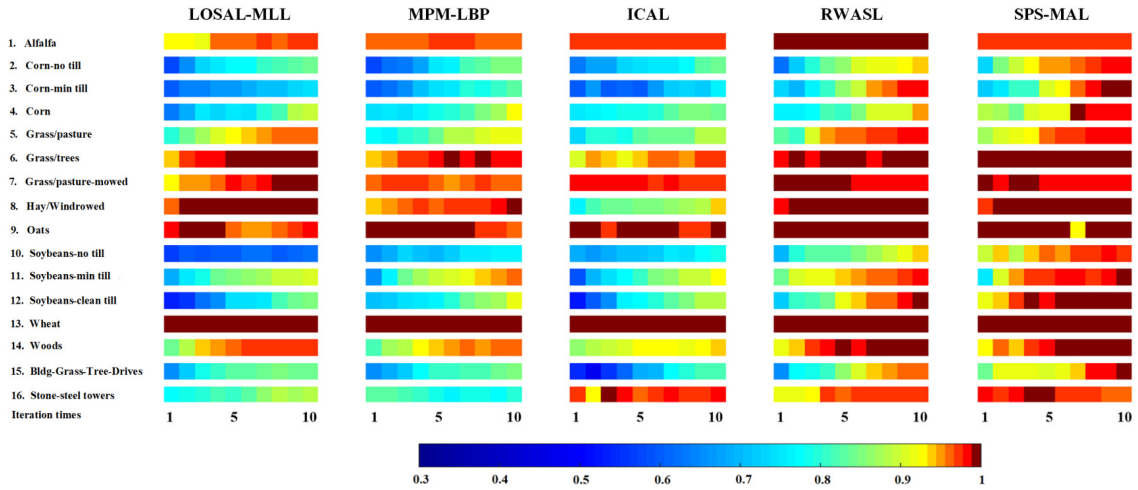


Fig. 7. CA of Indian Pines image along with ten iterations. The compared methods include LORSAL-MLL [15], MPM-LBP [49], ICAL [21], RWASL [17], and SPS-MAL.

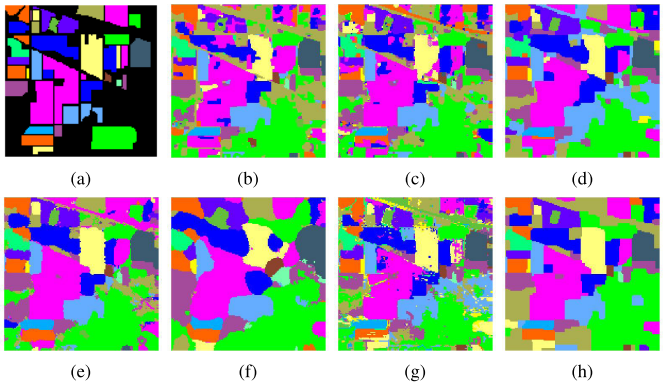


Fig. 8. Classification maps of Indian Pines image obtained by different methods with 198 samples (10 iterations). (a) is the Groundtruth and the compared methods are (b) LORSAL-MLL [15], (c) MPM-LBP [49], (d) ICAL [21], (e) RWASL [17], (f) MCMs [50], (g) HiFi [51], and (h) SPS-MAL, respectively.

due to the effective integration of diverse and complementary information provided by subpixel, pixel, and superpixel views. Moreover, the introduced MAL strategy can contribute to automatically selecting the most representative samples during each iteration process, leading to a further boost of classification performance.

To more clearly illustrate the class discrimination capability of different methods, Fig. 7 presents the comparison results of CA values along with the iterative AL process. In general, the SPS-MAL approach yields higher classification accuracies for most classes throughout the entire AL process, when compared to other AL-based methods. Specifically, obvious improvements for seven classes are obtained by applying the proposed method, including corn-no till, corn-min till, corn, soybean-no till, soybean-min till, soybean-clean till, and bldg-grass-tree-drives, especially at the last few iteration times. Furthermore, after 10 iteration times, the class-specific accuracies for all the 16 classes can reach above 94% by the SPS-MAL method, demonstrating the effectiveness of the proposed AL method.

Fig. 8 and Table I show the visual and numerical comparison results of Indian Pines image by different methods

TABLE I

CLASSIFICATION RESULTS (IN %) OF INDIAN PINES OBTAINED BY DIFFERENT METHODS WITH 198 SAMPLES BY 10 ITERATIONS (INITIAL TRAINING SAMPLE NUMBER FOR EACH CLASS IS 3 AND 15 MORE SAMPLES ARE ACTIVELY SELECTED PER ITERATION). THE COMPARED METHODS INCLUDE LORSAL-MLL [15], MPM-LBP [49], ICAL [21], RWASL [17], MCMS [50], HIFI [51], AND SPS-MAL

Class	LORSAL -MLL	MPM -LBP	ICAL	RWASL	MCMs	HiFi	SPS -MAL
1	96.09	95	96.74	**98.70**	88.7	98.04	95.87
2	76.52	79.96	85.53	91.37	82.42	83.86	**97.61**
3	62.70	75.41	63.01	97.36	81.02	88.37	**98.94**
4	85.06	89.58	93.63	92.78	95.23	86.08	**98.35**
5	91.04	88.92	81.57	**97.74**	92.11	88.16	97.60
6	98.81	98.16	99.01	99.58	95.48	95.93	99.70
7	98.93	95.71	96.43	98.21	94.64	96.07	97.14
8	**100**	99.29	97.51	99.96	97.87	96.36	**100**
9	97.50	95.00	68.00	**100**	**100**	99.50	**100**
10	50.67	66.97	82.71	91.02	81.57	88.5	**96.68**
11	88.43	94.33	91.64	97.21	92.13	89.66	**98.86**
12	79.29	88.48	97.40	98.45	78.72	86.19	**99.71**
13	99.71	99.56	99.56	**99.90**	98.05	99.02	99.71
14	96.62	93.92	99.30	**99.69**	91.84	96.17	99.55
15	78.58	78.81	74.84	94.69	85.23	88.91	**99.95**
16	81.61	71.61	98.71	95.48	**100**	98.60	94.41
OA(%)	82.78	87.27	88.85	96.38	88.58	90.16	**98.66**
AA(%)	86.35	88.17	89.10	97.01	90.94	92.46	**98.38**
Kappa(%)	80.16	85.36	87.24	95.87	87.07	88.81	**98.48**
Time cost(s)	5.29	212.13	12.90	5.82	61.74	269.28	26.30

with 10 iterations (198 samples in total), respectively. It is observed from Fig. 8 that both the RWASL and the SPS-MAL methods present a more similar visual impression to the groundtruth map. Meanwhile, SPS-MAL is more capable of edge-preserving and gets more similar structures with groundtruth. And the OA metric below each subfigure confirms it, mainly because of better classification performance in the classes of corn-no-till, Soybins-no till. In addition, it can be concluded from the class-by-class accuracies shown in Table I, that the introduced SPS-MAL scheme dominates the advantages in classifying the majority of classes (i.e., 9 out of 16 classes). Furthermore, the better classification results in terms of OA, AA, and κ coefficient indices obtained by the introduced multiview AL scheme demonstrate the overall superiority.

Another two data sets, "University of Pavia" and "Salinas," are used to evaluate the performance of different compared methods. For University of Pavia image, the initial sample

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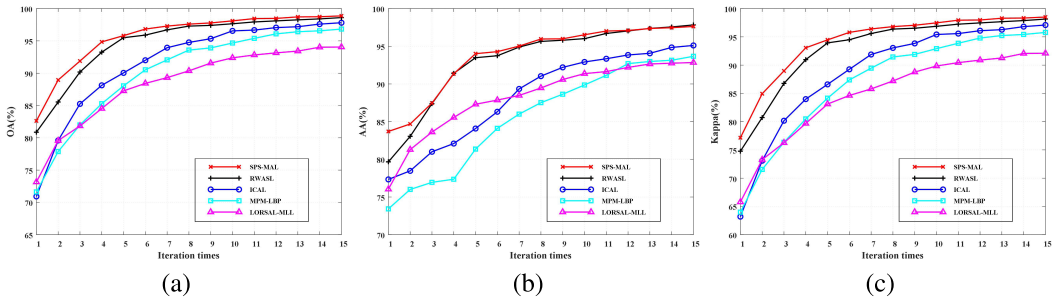


Fig. 9. Comparison classification results of University of Pavia image by different methods when the iteration times increase from 1 to 15, with 15 new samples selected per iteration. (a), (b), (c) are curves of OA, AA, and Kappa, respectively.

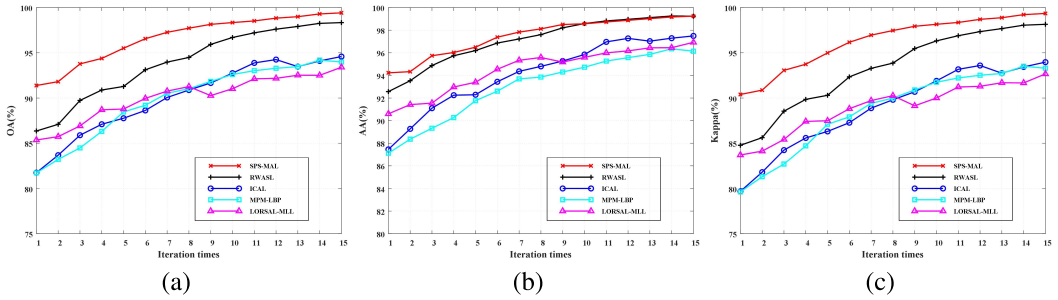


Fig. 10. Comparison classification results of Salinas image by different methods when the iteration times increase from 1 to 15, with 10 new samples selected per iteration. (a), (b), (c) are curves of OA, AA, and Kappa, respectively.

size is 27 where each class contains 3 samples, and 15 more samples are actively selected per iteration to maximize diversity or informativeness. For Salinas data sets, the initial sample size is 32, where each class contains 2 samples. During the iteration of AL process, 10 samples are selected for human labeling. All AL methods share the same initial samples.

Figs. 9 and 10 show the numerical comparison results of "University of Pavia" and "Salinas" images. Overall, it can be seen that the SPS-MAL method always generate higher accuracies than the compared AL methods, especially for the first few iteration times. This reflects the effectiveness of SPS-MAL method in combining multilevel information and selecting useful samples to boost discrimination capabilities. Though a competitive classification performance is obtained by the RWASL method when the number of training samples increases greatly, the obvious accuracy gap with the SPS-MAL method in the early stage demonstrate its drawback. The RWASL can generate good classification performance mainly due to the following reasons. One is that it takes advantage of random-walker algorithm and ensemble learning-based AL strategy (random-walker algorithm and extended random-walker algorithm), where disagreement analysis is adopted in sample selection. Another one is that it combines AL and semisupervised learning to further boost performance. When compared with the rest AL methods, including the LORSAL-MLL, MPM-LBP, and ICAL, more obvious advantage can be obtained by the proposed SPS-MAL method. Figs. 11 and 12 show the classification maps of "University of Pavia" and "Salinas," respectively. Overall, it can be observed that both the proposed SPS-MAL methods have more smooth and are more similar to the groundtruth map than the rest of the compared methods. It seems SPS-MAL performs a little worse than RWASL in some edges area, but RWASL has obviously misclassified some homogeneous areas. For example, the Grapes-untrained and Vineyard-untrained in Salinas image are quite difficult to be correctly classified. The visual comparison results show that the compared methods have generated varying degrees of obvious misclassification, and in contrast, a relative good discrimination results between these two classes can be generated by the SPS-MAL.

TABLE II

CLASSIFICATION RESULTS (IN %) OF UNIVERSITY OF PAVIA OBTAINED BY DIFFERENT METHODS WITH 177 SAMPLES BY 10 ITERATIONS (INITIAL TRAINING SAMPLE NUMBER FOR EACH CLASS IS 3 AND 15 MORE SAMPLES ARE ACTIVELY SELECTED PER ITERATION). THE COMPARED METHODS INCLUDE LORSAL-MLL [15], MPM-LBP [49], ICAL [21], RWASL [17], MCMS [50], HIFI [51], AND SPS-MAL

Class	LORSAL -MLL	MPM -LBP	ICAL	RWASL	MCMs	HiFi	SPS -MAL
1	94.15	97.41	97.51	**99.43**	90.91	87.10	98.95
2	96.66	98.12	99.29	99.84	99.15	96.91	99.77
3	75.10	73.42	78.30	**97.83**	93.66	75.75	85.41
4	89.31	93.29	**96.14**	71.78	74.19	78.06	93.72
5	**99.72**	91.41	**99.72**	96.73	99.85	89.35	99.59
6	84.00	96.56	99.25	99.19	84.03	87.56	**100**
7	90.49	66.58	71.37	**99.93**	68.82	86.77	99.77
8	91.38	93.24	95.18	**99.12**	92.60	85.59	95.90
9	**99.92**	99.01	99.50	98.38	39.90	**99.92**	95.64
OA(%)	92.72	94.68	96.55	97.40	91.24	90.44	**98.10**
AA(%)	91.19	89.89	92.92	95.80	82.57	87.44	**96.53**
Kappa(%)	90.29	92.92	95.42	96.53	88.26	87.29	**97.48**
Time cost(s)	29.82	2001.00	81.25	41.28	2720.43	45.59	184.61

The numerical comparison results (after 10 AL-based iteration times) of the two data sets ("University of Pavia" and "Salinas") are shown in Tables II and III, respectively. Overall, superior overall classification results in terms of three objective metrics are obtained by the proposed SPS-MAL method, when compared to the other methods. Besides, it is shown that better classification performance can be achieved by RWASL and the

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TABLE III

CLASSIFICATION RESULTS (IN %) OF SALINAS OBTAINED BY DIFFERENT METHODS WITH 132 SAMPLES BY 10 ITERATIONS (INITIAL TRAINING SAMPLE NUMBER FOR EACH CLASS IS 2 AND 10 MORE SAMPLES ARE ACTIVELY SELECTED PER ITERATION). THE COMPARED METHODS INCLUDE LORSAL-MLL [15], MPM-LBP [49], ICAL [21], RWASL [17], MCMs [50], HIFI [51], AND SPS-MAL

Class	LORSAL -MLL	MPM -LBP	ICAL	RWASL	MCMs	HiFi	SPS -MAL
1	99.64	98.18	99.86	99.97	98.28	98.74	**100**
2	**100**	98.97	99.92	**100**	93.30	98.67	99.98
3	98.24	96.95	99.05	**100**	95.14	96.97	99.97
4	99.68	98.46	99.59	**99.95**	96.70	97.33	98.95
5	99.23	99.01	99.26	**99.45**	80.45	98.37	99.06
6	99.99	99.10	99.95	**100**	94.29	97.83	99.94
7	99.92	99.78	99.86	**100**	99.26	98.90	99.62
8	83.34	89.65	91.49	92.28	97.01	83.12	**98.19**
9	99.67	99.54	99.73	**100**	95.63	99.85	**100**
10	89.33	93.98	97.08	**99.14**	86.90	82.49	97.51
11	91.21	83.89	90.12	**99.65**	97.28	93.42	98.23
12	99.95	98.81	**100**	**100**	84.15	99.79	99.58
13	99.29	98.22	97.23	**99.42**	95.52	97.07	98.43
14	96.95	94.21	97.33	**99.68**	91.73	89.79	93.82
15	64.9	72.25	63.94	88.25	85.53	80.58	**93.89**
16	99.01	94.83	99.19	99.61	88.43	87.33	**99.86**
OA(%)	90.72	92.61	92.75	96.70	92.68	91.36	**98.34**
AA(%)	95.02	94.74	95.85	**98.59**	92.48	93.77	98.57
Kappa(%)	89.66	91.75	91.90	96.32	91.84	90.38	**98.15**
Time cost(s)	22.38	1141.5	62.03	25.47	2604.45	342.98	129.30

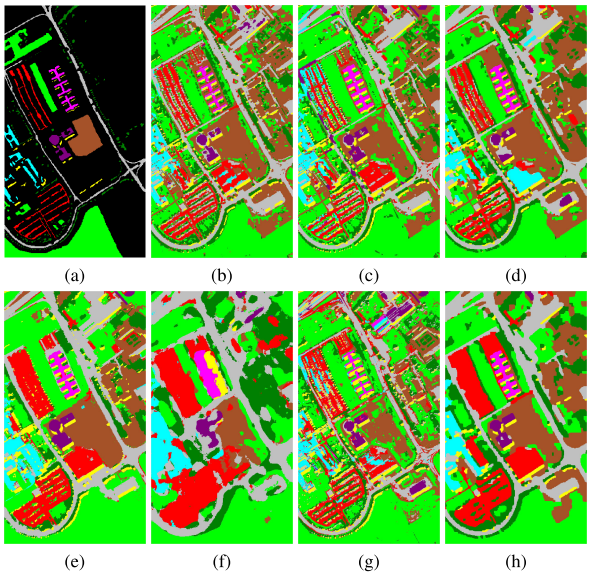


Fig. 11. Classification maps obtained of University of Pavia image by different methods with 177 samples (10 iterations). (a) is the Groundtruth and the compared methods are (b) LORSAL-MLL [15], (c) MPM-LBP [49], (d) ICAL [21], (e) RWASL [17], (f) MCMs [50], (g) HiFi [51], and (h) SPS-MAL, respectively.

proposed SPS-MAL methods than the supervised classification approaches, i.e., MCMs and HiFi. This should be owing to the effective AL strategies to select more representative and informative samples than the random sample selection manner. In addition, the classification performance by two supervised methods (i.e., MCMs and HiFi) is competitive with two of the classical AL methods for HSIs (i.e., LORSAL-MLL and MPM-LBP). This demonstrates that effective multiinformation fusion strategy can contribute to a boost of classification accuracies. In the proposed SPS-MAL method, the benefits brought by multilevel information fusion and effective AL strategy are simultaneously exploited, and thus, overall superior performance is obtained.

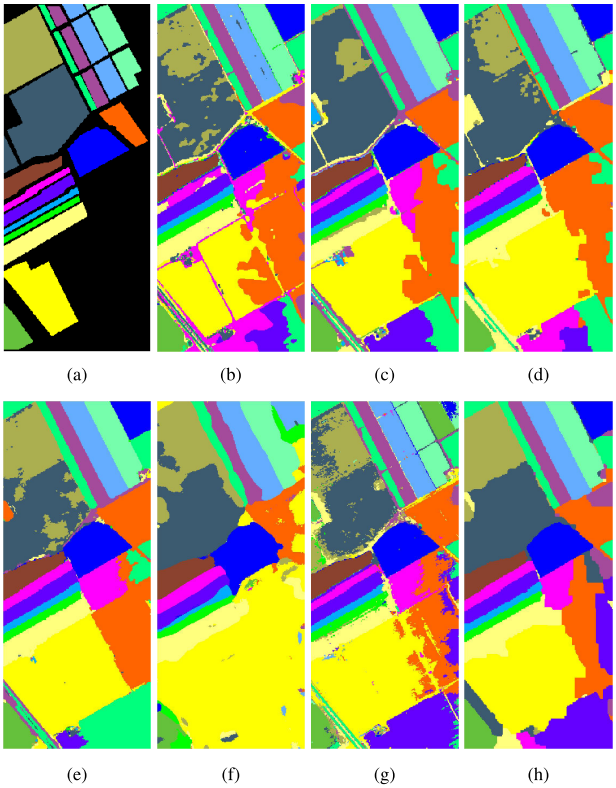


Fig. 12. Classification maps of Salinas image obtained by different methods with 132 samples (10 iterations). (a) is groundtruth and the compared methods are (b) LORSAL-MLL [15], (c) MPM-LBP [49], (d) ICAL [21], (e) RWASL [17], (f) MCMs [50], (g) HiFi [51], and (h) SPS-MAL, respectively.

D. Comparison Of Different AL Strategies

To compare the performance of different AL strategies in the proposed classification framework, we have replaced BT with other commonly used strategies, i.e., max entropy criterion (ENTROPY) [52], mutual information criterion (MI) [53], MBT, and random sampling (RS). Specifically, the AL strategies mentioned above are used in the same AL framework proposed in this article. Take the RS strategy as an example, based on the proposed SPS-MAL scheme, the newly added samples by the RS are selected via random sampling in the candidate set U, where samples are generated by the coarse sample selection stage (i.e., samples with contrary labels on different views). The experiments in this part are conducted on test data of Pavia University, which contains rich homogeneous and heterogeneous areas. Fig. 13 shows the curves of OA, AA, and Kappa metric when iteration time increases from 1 to 15. The detailed numerical results of Pavia University are presented in Table IV. It can be seen that BT always ranks the first along with the iteration process, in terms of OA, AA, and κ coefficient. Among the rest compared strategies, both ENTROPY and MI focus on the most complex areas during sample selection, i.e., areas in multiclass boundaries, neglecting the balance between each class. RS strategy generates moderate performance because RS introduces more balanced samples than ENTROPY and MI, while less informative samples compared with MBT and BT. The MBT is a modified version of BT, which is designed to

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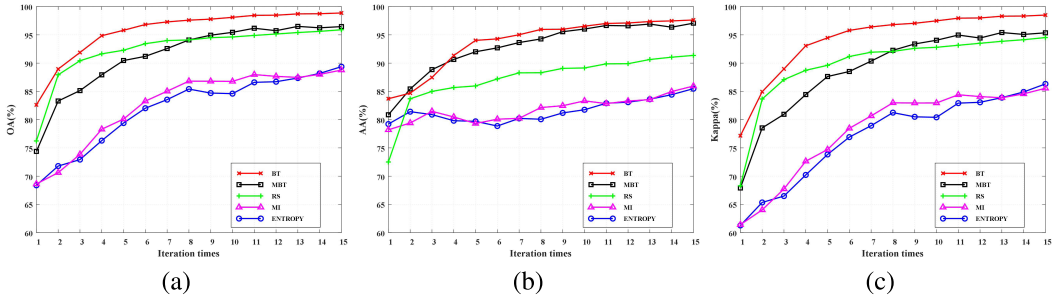


Fig. 13. Comparison classification results of Pavia image by different ranking methods when the iteration times increase from 1 to 15, with 15 new samples selected per iteration. (a), (b), (c) are curves of OA, AA, and Kappa, respectively.

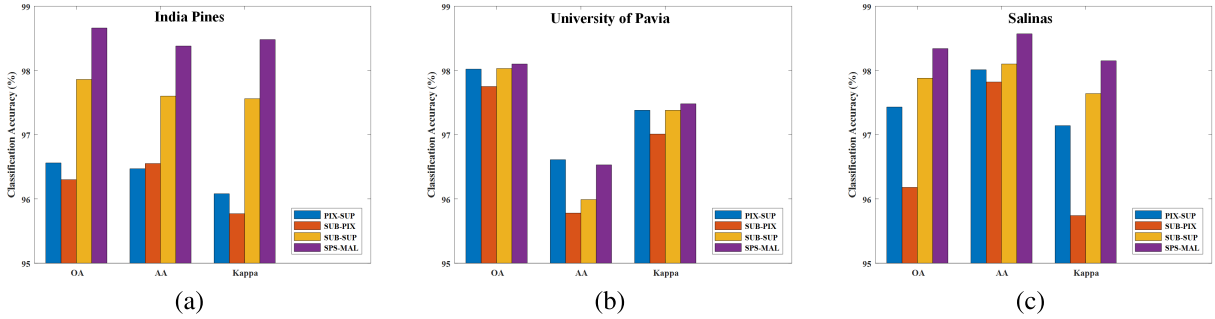


Fig. 14. Classification results with part of multiple views. The pixel-view, subpixel-view, and superpixel-view is abbreviated to PIX, SUB, and SUP, respectively. (a), (b), (c) are classification results of Indian Pines, University of Pavia, and Salinas images, respectively.

TABLE IV

CLASSIFICATION RESULTS (IN %) OF PAVIA UNIVERSITY OBTAINED BY DIFFERENT RANKING METHODS WITH 177 TRAINING SAMPLES IN TOTAL. THE COMPARED RANKING METHODS INCLUDE ENTROPY [52], MI [53], MBT [15], RS, AND BT [14]

Class	ENTROPY	MI	MBT	RS	BT
1	98.44	98.55	96.39	96.64	**98.95**
2	78.85	83.97	94.85	**99.99**	99.77
3	66.13	70.34	**87.39**	77.45	85.41
4	86.64	85.73	**95.66**	72.95	93.72
5	98.91	**99.64**	99.58	99.39	99.59
6	98.76	95.94	99.12	99.77	**100**
7	19.15	17.09	**99.95**	66.65	99.77
8	94.07	**96.15**	92.09	92.40	95.90
9	94.83	94.72	**99.22**	97.05	95.64
OA(%)	84.60	86.79	95.45	94.63	**98.10**
AA(%)	81.75	82.46	96.03	89.14	**96.53**
Kappa(%)	80.41	82.94	94.05	92.80	**97.48**

introduce diversity in sample selection. However, experiments demonstrate the BT performs better than MBT here. Thus, BT is adopted as the sample selection strategy in the proposed method.

E. Time Cost Comparison of Different Classification Methods

Regarding time costs, we first state our software and hardware equipment for experiments. The hardware equipment consists of an Intel Core i7-6850K CPU with 6 cores, 32-GB RAM. These codes were implemented under the Windows10 operating system and in MATLAB17 software. The time costs by different methods are presented in Tables I–III. Among them, MCM is the most time-consuming algorithm, which is caused by the time-costing process of DL and training. Besides, the time cost for the HiFi depends not only on data size but also on the number of multiple classifiers for ensemble learning. The AL algorithm, MPM-LBP gets the main time cost due to the adopted loop belief propagation model. In contrast, the LORSAL-MLL, ICAL, and RWASL cost less time due to the fact that the sample selection process and classification process are of higher efficiency. Though the proposed method generates a moderate time cost due to the processes of multiview generation and MAL, the running time can be reduced by the parallel computing strategy. This is reasonable for the proposed SPS-MAL method, since the processes of view generation, classifier training, and Markov random field-based optimization for each view can be carried out independently. Furthermore, it should be noted that the training sample labeling process in the AL framework is a manual process in practice, which requires much time and effort. With the proposed method, less samples are needed to generate higher classification accuracies, which, overall, can lead to a lower time cost.

F. Effect of Combining Multiple Views for Classification

In this section, the complementary properties of the aforementioned multiple views and the necessity of combining multiple views are investigated. Based on the proposed MAL framework, difference maps need to be preformed to generate a sample candidate set. Therefore, at least two of the views should be considered. Here, the AL-based classification methods by combining only two of the generated views are compared. These methods are named by SUB-PIX (subpixel-level and pixel-level views), PIX-SUP (pixel-level and superpixel-level views), and SUB-SUP (subpixel-level and superpixel-level views). Fig. 14 presents the detailed results

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experimented on three data sets after 10 iterations, in terms of OA, AA, and Kappa coefficients.

In general, it can be observed from Fig. 14 that the SPS-MAL method always obtains the best results in terms of OA- and AA-based classification accuracies, when three of the considered views were combined. This is expected because subpixel-level, pixel-level, and superpixel-level views provide diverse and complementary information, which can be jointly exploited with the proposed MAL schemes for better classification results. The AA value of the University of Pavia by SPS-MAL method presents slightly a lower index than PIX-SUP due to the fact that the spectral discrimination information and spatial structure information mainly contribute to increasing classification performance, when compared to the subpixel-based abundance information. However, the higher OA and kappa coefficients indicate the effectiveness of the integration of three levels of views in this data set. Moreover, it is worth noticing that big boost can be generated when comparing the SPS-MAL with the two-view-based AL classifier for the data sets of Indian Pines and Salinas. Overall, the experimental results can demonstrate the effectiveness of the proposed MAL scheme in the improvement of classification results, which should benefit from the integration of subpixel-level, pixel-level, and superpixel-level views.

IV. Conclusion

In this article, we propose a novel SPS-MAL method for hyperspectral classification. The experimental results on three HSIs demonstrate the superiority in classification performance of our method when compared to state-of-the-art AL methods and supervised methods. On the one hand, the diverse and complementary information provided by subpixel-level, pixel-level, and superpixel-level views are jointly exploited by the SPS-MAL method, leading to an improvement in discrimination capability between different classes. On the other hand, a new multiview active leaning strategy via joint difference analysis on multiple views and joint posterior probability estimation is introduced, contributing to the effective selection of the most representative samples. Therefore, the overall classification accuracies can be effectively boosted while the required necessary training samples can be significantly reduced. However, there still exist some areas for improvement. For example, it is assumed that the manually labeled samples used for training are accurate, which may be invalid in practical applications. Therefore, one of the main research focus in the future will be to investigate the effective approaches to identify the most likely wrongly labeled samples and correct them with the proposed MAL framework. Another focus of the future work will be laid on the development of semisupervised multiview learning methods for HSIs, aiming at taking advantage of the large volumes of unlabeled samples to further improve classification performance.

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